

Exercise 2: (8 points)

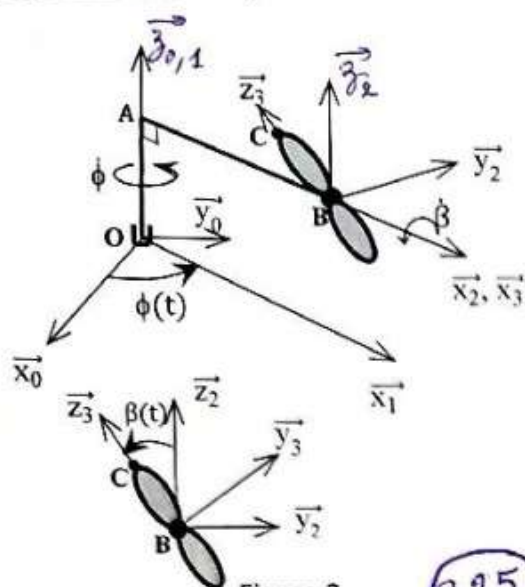


Figure 2

$$1. \vec{\Omega}_{3/R2}^0 = \vec{\Omega}_{3/R2}^2 + \vec{\Omega}_{2/R2}^1 + \vec{\Omega}_{1/R2}^0 = \dot{\beta} \vec{x}_2 + \vec{0} + \dot{\phi} \vec{z}_2 = \begin{Bmatrix} \dot{\beta} \\ 0 \\ \dot{\phi} \end{Bmatrix}_{R2}$$

$$2. \overline{V^0(B)}_{/R2} = \frac{d^0 \vec{OB}}{dt}_{/R2} = \frac{d^2 \vec{OB}}{dt}_{/R2} + \vec{\Omega}_{2/R2}^0 \wedge \vec{OB}_{/R2} = \begin{Bmatrix} 0 \\ 0 \\ 0 \end{Bmatrix}_{R2} + \begin{Bmatrix} 0 \\ 0 \\ \dot{\phi} \end{Bmatrix}_{R2} \wedge \begin{Bmatrix} b \\ 0 \\ a \end{Bmatrix}_{R2} = \begin{Bmatrix} 0 \\ b\dot{\phi} \\ 0 \end{Bmatrix}_{R2}$$

$$\overline{\gamma^0(B)}_{/R2} = \frac{d^0 \overline{V^0(B)}}{dt}_{/R2} = \frac{d^2 \overline{V^0(B)}}{dt}_{/R2} + \vec{\Omega}_{2/R2}^0 \wedge \overline{V^0(B)}_{/R2}$$

$$= \begin{Bmatrix} 0 \\ 0 \\ 0 \end{Bmatrix}_{R2} + \begin{Bmatrix} 0 \\ 0 \\ \dot{\phi} \end{Bmatrix}_{R2} \wedge \begin{Bmatrix} 0 \\ b\dot{\phi} \\ 0 \end{Bmatrix}_{R2} = \begin{Bmatrix} -b\dot{\phi}^2 \\ 0 \\ 0 \end{Bmatrix}_{R2}$$

$$3. \overline{V^0(C)}_{/R2} = \overline{V^0(B)}_{/R2} + \vec{\Omega}_{3/R2}^0 \wedge \vec{BC}_{/R2} = \begin{Bmatrix} 0 \\ b\dot{\phi} \\ 0 \end{Bmatrix}_{R2} + \begin{Bmatrix} \dot{\beta} \\ 0 \\ \dot{\phi} \end{Bmatrix}_{R2} \wedge \begin{Bmatrix} 0 \\ -r \sin \beta \\ r \cos \beta \end{Bmatrix}_{R2} = \begin{Bmatrix} \dot{\phi} r \sin \beta \\ b\dot{\phi} - \dot{\beta} r \cos \beta \\ -\dot{\beta} r \sin \beta \end{Bmatrix}_{R2}$$

$$\overline{\gamma^0(C)}_{/R2} = \overline{\gamma^0(B)}_{/R2} + \frac{d^0 \vec{\Omega}_3}{dt}_{/R2} \wedge \vec{BC}_{/R2} + \vec{\Omega}_{3/R2}^0 \wedge (\vec{\Omega}_{3/R2}^0 \wedge \vec{BC}_{/R2})$$

$$\text{Avec } \frac{d^0 \vec{\Omega}_3}{dt}_{/R2} = \frac{d^2 \vec{\Omega}_3}{dt}_{/R2} + \vec{\Omega}_{2/R2}^0 \wedge \vec{\Omega}_{3/R2}^0 = \begin{Bmatrix} 0 \\ 0 \\ 0 \end{Bmatrix}_{R2} + \begin{Bmatrix} 0 \\ 0 \\ \dot{\phi} \end{Bmatrix}_{R2} \wedge \begin{Bmatrix} \dot{\beta} \\ 0 \\ \dot{\phi} \end{Bmatrix}_{R2} = \begin{Bmatrix} 0 \\ \dot{\beta} \dot{\phi} \\ 0 \end{Bmatrix}_{R2}$$

$$\frac{d^0 \vec{\Omega}_3}{dt}_{/R2} \wedge \vec{BC}_{/R2} = \begin{Bmatrix} 0 \\ \dot{\beta} \dot{\phi} \\ 0 \end{Bmatrix}_{R2} \wedge \begin{Bmatrix} 0 \\ -r \sin \beta \\ r \cos \beta \end{Bmatrix}_{R2} = \begin{Bmatrix} \dot{\beta} \dot{\phi} r \cos \beta \\ 0 \\ 0 \end{Bmatrix}_{R2}$$

$$\vec{\Omega}_{3/R2}^0 \wedge \vec{BC}_{/R2} = \begin{Bmatrix} \dot{\beta} \\ 0 \\ \dot{\phi} \end{Bmatrix}_{R2} \wedge \begin{Bmatrix} 0 \\ -r \sin \beta \\ r \cos \beta \end{Bmatrix}_{R2} = \begin{Bmatrix} \dot{\phi} r \sin \beta \\ -\dot{\beta} r \cos \beta \\ -\dot{\beta} r \sin \beta \end{Bmatrix}_{R2}$$

$$\overline{\Omega}_{3/R2}^0 \wedge (\overline{\Omega}_{3/R2}^0 \wedge \overline{BC}_{/R2}) = \begin{Bmatrix} \dot{\beta} \\ 0 \\ \dot{\Phi} \end{Bmatrix}_{R2} \wedge \begin{Bmatrix} \Phi r \sin \beta \\ -\dot{\beta} r \cos \beta \\ -\dot{\beta} r \sin \beta \end{Bmatrix}_{R2} = \begin{Bmatrix} \dot{\beta} \dot{\Phi} r \cos \beta \\ \Phi^2 r \sin \beta + \dot{\beta}^2 r \sin \beta \\ -\dot{\beta}^2 r \cos \beta \end{Bmatrix}_{R2} = \begin{Bmatrix} \dot{\beta} \dot{\Phi} r \cos \beta \\ (\Phi^2 + \dot{\beta}^2) r \sin \beta \\ -\dot{\beta}^2 r \cos \beta \end{Bmatrix}_{R2} \quad \text{0,5}$$

Finalemment,

$$\overline{\gamma^0(C)}_{/R2} = \begin{Bmatrix} -\frac{1}{2}\dot{\Phi}^2 \\ 0 \\ 0 \end{Bmatrix}_{R2} + \begin{Bmatrix} \dot{\beta} \dot{\Phi} r \cos \beta \\ 0 \\ 0 \end{Bmatrix}_{R2} + \begin{Bmatrix} \dot{\beta} \dot{\Phi} r \cos \beta \\ (\Phi^2 + \dot{\beta}^2) r \sin \beta \\ -\dot{\beta}^2 r \cos \beta \end{Bmatrix}_{R2} = \begin{Bmatrix} -\frac{1}{2}\dot{\Phi}^2 + 2\dot{\beta} \dot{\Phi} r \cos \beta \\ (\Phi^2 + \dot{\beta}^2) r \sin \beta \\ -\dot{\beta}^2 r \cos \beta \end{Bmatrix}_{R2} \quad \text{0,5}$$